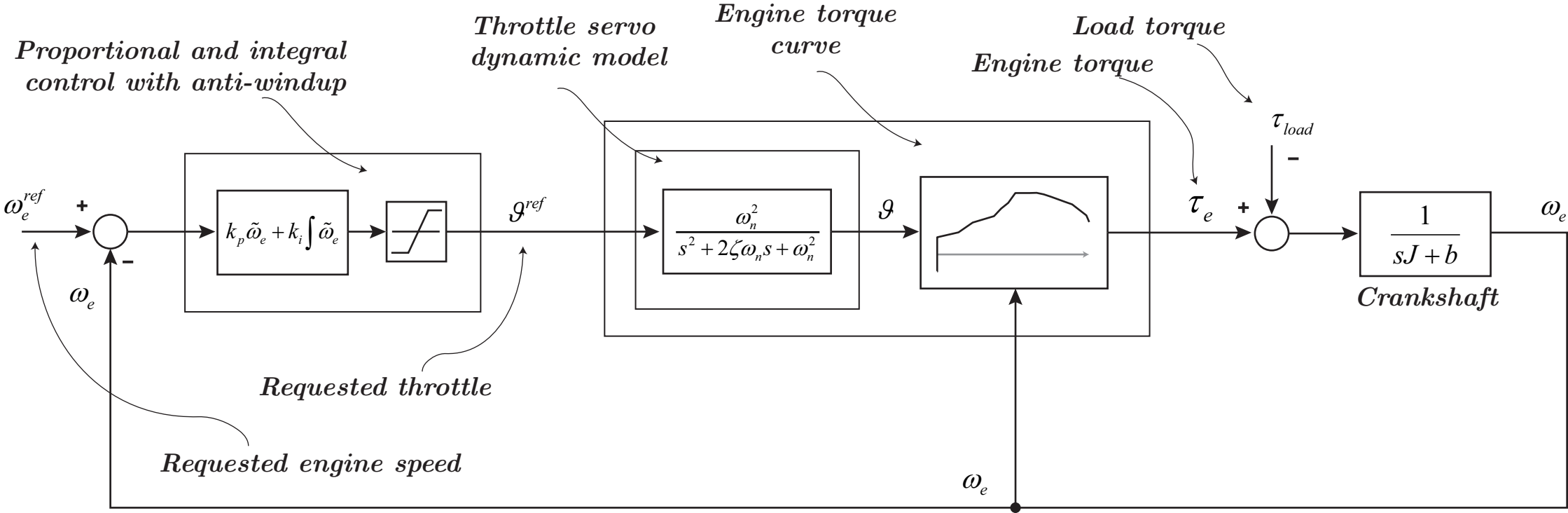
Proportional and integral control with anti-windup
Throttle servo dynamic model
Engine torque curve
Load torque
Engine torque
ω_e^{ref}
+
−
ω_e
$k_p\tilde{\omega}_e + k_i\int\tilde{\omega}_e$
ϑ^{ref}
$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
ϑ
τ_e
τ_{load}
$\frac{1}{sJ+b}$
Crankshaft
ω_e
Requested throttle
Requested engine speed
ω_e